1. $\{f_n\}_n$ are $\mathbb{R}$ valued Borel-measurable on Ω

$\mathcal{F}$ is σ-field containing Ω.

Consider $f \underline{=}$.

$$f(\omega) = \sup_n f_n(\omega) = \sup\{f_1(\omega), f_2(\omega), \dots\}$$

$$f^{-1}((-\infty, x]) \in \mathcal{F} \quad \forall\ x \in \mathbb{R}$$

WE claim:

$$\{\omega : f(\omega) \le x\} = \bigcap_{i=1}^{\infty} \{\omega : f_i(\omega) \le x\}$$

$$\bigcap_{i=1}^{\infty} f_i^{-1}((-\infty, x])$$

Consider $t \in \{\omega : f(\omega) \le x\}$

$\Leftrightarrow$ $f(t) \le x$

$\Leftrightarrow$ $\sup_n f_n(t) \le x$

$f_n(t) \le x \quad \forall\ n$

$\Leftrightarrow$

$\Leftrightarrow$ $t \in \{\omega : f_n(\omega) \le x\} \quad \forall n$

$\Leftrightarrow$ $t \in \bigcap \{\omega : f_n(\omega) \le x\}$.

Now $\{\omega : f(\omega) \le x\} = \bigcap \{\omega : f_i(\omega) \le x\}$

So, Now, $f^{-1}((-\infty, x]) = \bigcap_{i=1}^{\infty} f_i^{-1}((-\infty, x])$

Now: $\because$ each $f_i^{-1}((-\infty, x]) \in \mathcal{F} \quad \forall x$

($\because$ f_i's are Borel measurable)

$\Rightarrow$ $\bigcap_{i=1}^{\infty} f_i^{-1}((-\infty, x]) \in \mathcal{F}$ $\forall x$ (property of σ-field)

hence $f^{-1}((-\infty, x]) \in \mathcal{F} \quad \forall x \in \mathbb{R}$

thus f is Borel-measurable

Now, for $\omega \longmapsto \inf_n f_n(\omega)$

Take

try proving $f^{-1}([x, \infty)) \in \mathcal{F} \quad \forall x \in \mathbb{R}$

& Also claim: $f^{-1}([x, \infty)) = \bigcap_{i=1}^{\infty} f_i^{-1}([x, \infty))$

(b) Consider

$$f(\omega) = \lim_{n\to\infty} \sup_n f_n(\omega)$$

$$= \lim_{n\to\infty} \sup\{f_k(\omega) : k \geq n\}$$

$$S_n = \sup\{f_k(\omega) : k \geq n\}$$

$$S_n \geq S_{n+1} \quad \forall n$$

So. $\lim \sup_n f_n(\omega) = \inf_n \left(\sup\{f_k(\omega) : k \geq n\}\right)$

consider. set $\sup\{f_k(\omega) : k \geq n_0\}$

This can be mapped to $g_i(\omega)$

s.t. $g_i(\omega) = f_{i+n_0-1}(\omega) \quad \forall i \geq 1$

hence

$\sup\{f_k(\omega) : k \geq n_0\} = \sup\{g_k(\omega) : k \geq 1\}$

$= \sup_k g_k(\omega)$

Now. define this

$g'_{n_0}(\omega) = \sup_k g_k(\omega)$

From above result of we have partial $g'_{n_0}(\omega)$ is measurable

$\forall\ n_0 \geq 1$

Now. $f(\omega) = \lim\sup_n f_n(\omega) = \inf_n(g'_n(\omega))$

Also we proved above $\omega \to \inf_n(g'_n(\omega))$ would be measurable given that g'_n are measurable

hence $f(\omega)$ is measurable

Similarly we prove for $f(\omega) = \lim_{n\to\infty} \inf_n f_n(\omega)$

2. (a) We know that F_x is non decreasing

Now, $\lim_{y \to x^-} F_x(y) = \inf(\{F_x(y) : y < x\})$

$\lim_{y \to x^+} F_x(y) = \sup(\{F_x(y) : y > x\})$

Now, Take $y_1 < x$ & $y_2 > x$

then $F_x(y_1) \leq F_x(y_2)$ $\forall\, y_1 < x$ & $\forall y_2 > x$

$\Rightarrow \sup(\{F_x(y_1) : y_1 < x\}) \leq F_x(y_2)$ $\forall y_2 > x$

$\Rightarrow$

$\sup(\{F_x(y_1) : y_1 < x\}) \leq \inf(\{F_x(y_2) : y_2 > x\})$

$\lim_{y \to x^-} F_x(y) \leq \lim_{y \to x^+} F_x(y)$

Now,

f discontinuing iff.

$\lim_{y \to x^-} F_x(y) = \sup(\{F_x(y_1) : y_1 < x\})$

$\inf(\{F_x(y_2) : y_2 > x\}) = \lim_{y \to x^+} F_x(y)$

Now consider set of all discontinuities as D

define $h : D \to \mathbb{Q}$ s.t.

$h(x)$ = a rational strictly b/w $\lim_{y \to x^-} F_x(y)$ & $\lim_{y \to x^+} F_x(y)$

Claim: h is injective

Consider WLOG $x_1 < x_2 \in D$

then. $\lim_{y \to x_1} F_x(x_1) < h(x_1) < \lim_{y \to x_1^+} F_x(x_1) \leq \lim_{y \to x_2^-} F_x(x_2)$

$< h(x_2) <$

$\lim_{y \to x_2^+} F_x(x_2)$

hence $h(x_1) \neq h(x_2)$ $\forall x_1$ & x_2

$\therefore$ h is injective

$\therefore$ h is injective $\therefore$ $|D| \leq |\mathbb{Q}| = |\mathbb{N}|$

hence D is countable.

(b) $D = \{x_1, x_2, \dots\}$ is set of discontinuities of F_X

$$F_X(x_j) - F_X(x_j-) = \text{Jump at } x_j$$

$F_d : \mathbb{R} \to \mathbb{R}$ by

$$F_d(x) = \sum_{x_j \in D} \left(F_X(x_j) - F_X(x_j-)\right) 1_{(-\infty, x]}(x_j)$$

$x \in \mathbb{R}$

(i) $\because$ $F_X(x_j) \geq F_X(x_j-)$ ($\because$ F_X is non-decreasing)

hence $F_X(x_j) - F_X(x_j-) > 0 \quad \forall x_j$

Also $1_{(-\infty, x_j]}(x) \geq 0 \quad \forall x_j$

hence $\left(F_X(x_j) - F_X(x_j-)\right) 1_{(-\infty, x_j]}(x) \geq 0$

$\forall x_j$

hence $F_d(x) = \sum_{x_j \in D} \left(F_X(x_j) - F_X(x_j-)\right) 1_{(-\infty, x_j]}(x)$

≥ 0

So $F_d(x) \geq 0$

$F_X(x_j) = P \circ X^{-1}((-\infty, x_j])$

So, $F_X(x_j-) = P \circ X^{-1}\left(\bigcup_{i=1}^{\infty} (-\infty, x_i]\right)$ apply $\lim$

$= P \circ X^{-1}((-\infty, x_i))$

hence $F_X(x_j) - F_X(x_j-) = P \circ X^{-1}((-\infty, x_j]) - P \circ X^{-1}((-\infty, x_j))$

$= P \circ X^{-1}(\{x_j\}) > 0$

$$F_d(x) = \sum_{x_j \in D} P \circ X^{-1}(\{x_j\})\, 1_{(-\infty, x]}(x_j) = \sum_{\substack{x_j \in D \\ \& x_j \leq x}} P \circ X^{-1}(\{x_j\})$$

$$= P_0 X^{-1}\left(\{x_i : x_i \le x \ \& \ x_i \in D\}\right) \quad -(1)$$

Now,

$$\because \quad \{x_i : x_i \le x \ \& \ x_i \in D\} \subseteq (-\infty, x] \quad \forall x$$

how.

$$P_0 X^{-1}\left(\{x_i : x_i \le x \ \& \ x_i \in D\}\right) \le P_0 X^{-1}\left((-\infty, x]\right) \quad \forall x$$

$$\Rightarrow \quad F_d(x) \le F_X(x) \quad \forall x$$

(ii) Consider x_1, x_2 s.t. $x_1 < x_2$

then.

$$F_d(x_2) - F_d(x_1) = P_0 X^{-1}\left(\{x_i : x_i \in D \ \& x_1 < x_i \le x_2\}\right)$$

≥ 0 ($\because P_0 X^{-1}(m) \ge 0 \ \forall m \in \mathcal{B}_{\mathbb{R}}$)

$F_d(x)$ is non-decreasing.

$F_d(x+) =$

$$= \lim_{y \to x+} F_d(y) = \inf\{F_d(y) : y > x\} \quad (\because F_d \text{ is non-decreasing})$$

$$= \inf\{P_0 X^{-1}(\{x_i : x_i \le y \ \& x_i \in D\} : y > x\}$$

$$= P_0 X^{-1}(\{x_i : x_i \le x \ \& \ x_i \in D\})$$

$$= F_d(x) \quad \text{sup. of } \{P_0 X^{-1}\}$$

hence F_d is right continuous

(iii) $\lim_{x \to -\infty} F_d(x) = P_0 X^{-1}(\{\phi\}) = 0$

$\lim_{x \to +\infty} F_d(x) = P_0 X^{-1}(D)$

$\because P_0 X^{-1}$ is clearly defined for each $A \in \mathcal{B}_{\mathbb{R}}$ & $D \in \mathcal{B}_{\mathbb{R}}$ hence $P_0 X^{-1}(D)$ is defined.

hence limit exists

(c) (ii) $F_c : \mathbb{R} \to \mathbb{R}$ $\quad F_c(x) = F_X(x) - F_d(x) \quad x \in \mathbb{R}$

$$F_c(x) = P \circ X^{-1}\left((-\infty, x]\right) - P \circ X^{-1}\underbrace{\left(\{x_j : x_j \le x \,\&\, x_j \in D\}\right)}_{A_x}$$

$$= P \circ X^{-1}\left((-\infty, x] \setminus A_x\right) \le P \circ X^{-1}\left((-\infty, x]\right)$$

Also $\because A_x \subseteq D$ hence A_x is countable

Now. $\quad \phi \subseteq (-\infty, x] \setminus A_x$

$$0 = P \circ X^{-1}(\phi) \le F_c(x)$$

(ii) F_c is non-decreasing.

consider $x_1 < x_2$

$$F_c(x_2) - F_c(x_1) = \left(F_X(x_2) - F_X(x_1)\right) - \left(F_d(x_2) - F_d(x_1)\right)$$

$$= P \circ X^{-1}\left((x_1, x_2]\right) - P \circ X^{-1}\underbrace{\left(A_{x_2} \setminus A_{x_1}\right)}_{\ge 0}$$

$\because A_{x_2} \setminus A_{x_1} \subseteq (x_1, x_2]$ (By defn)

Hence F_c is non-decreasing.

Now.

$$F_c(x+) = F_X(x+) - F_d(x+)$$

$$= F_X(x) - F_d(x) = F_c(x)$$

Now

$$F_c(x-) = F_X(x-) - F_d(x-)$$

$$F_c(x) = F_X(x) - F_d(x)$$

$$F_c(x) - F_c(x-) = \left(F_X(x) - F_X(x-)\right) - \left(F_d(x) - F_d(x-)\right)$$

if x is pt. of discontn. then.

$$\left(F_X(x) - F_X(x-)\right) = P \circ X^{-1}(\{x\}) = F_d(x) - F_d(x-)$$

If x is pt. of continuity then.

$$\left(F_X(x) - F_X(x-)\right) = 0$$

Also $\quad F_d(x) - F_d(x-) = P \circ X^{-1}(\phi) = 0$

hence $F_1(x)$ is continuous.

(iii) $\lim_{x \to -\infty} F_1(x) = 0 - 0 = 0$

$\lim_{x \to \infty} F_1(x) = 1 - (P \circ X^{-1})(D)$ hence exists.

(d) $\alpha = \lim_{x \to \infty} F_d(x)$ $\beta = \lim_{x \to \infty} F_1(x)$

(i) ✓

(ii) ✓

(iii)

(iv) ✓ $\beta = 0$

(v)

(vi)

$\therefore x_i$ is discontinuous here.

for $x_i \in D$ $p(X = x_i) = \frac{1}{\alpha} \cdot \left(F_d(x_i) - F_d(x_{i-}) \right) = \frac{(P \circ X^{-1})(\{x_i\})}{(P \circ X^{-1})(D)}$

$\therefore \{x_i\} \subseteq D$

$0 < p(X = x_i) \leq 1$

Also $\sum p(X = x_i) =$

3. $\mathcal{B}_{\mathbb{R}} = \mathcal{B}_{\bar{\mathbb{R}}} \cap \mathbb{R}$ ← σ-field

$\mathbb{R} \subseteq \mathcal{B}_{\mathbb{R}}$ $\{ B \in \mathcal{B}_{\bar{\mathbb{R}}} \mid B \cap \mathbb{R} \in \mathcal{B}_{\mathbb{R}} \}$

Consider.

$E = (E \cap \mathbb{R}) \cup (E \cap \{-\infty, +\infty\})$

Claim: Any set is open in $\mathbb{R}$ iff it can be written as $O \cap \mathbb{R}$ for some open set $\in \bar{\mathbb{R}}$

Using good sets.

$\mathcal{B}_{\bar{\mathbb{R}}} \cap \mathbb{R} \subseteq \mathcal{B}_{\mathbb{R}}$

Also $A \in \mathcal{B}_{\mathbb{R}} \subseteq \mathcal{B}_{\bar{\mathbb{R}}}$ so, $A \in \mathcal{B}_{\bar{\mathbb{R}}}$

Hence $A \cap \mathbb{R} = A \in \mathcal{B}_{\mathbb{R}}$

Now. $\mathcal{B}_{\mathbb{R}} \subseteq \mathcal{B}_{\bar{\mathbb{R}}} \cap \mathbb{R}$

4. $A = \bigcup_n A_n$ i.e. $A_n \uparrow A$

$A_n \in \mathcal{F}$

$\mathbb{1}_{A_1} \leq \mathbb{1}_{A_2} \leq \dots \mathbb{1}_A$

consider $\mathbb{1}_{A_k}(\omega) = 1 \Rightarrow \omega \in A_k \Rightarrow \omega \in A_{k+1}$ $(\because A_k \subseteq A_{k+1})$

$\Rightarrow \mathbb{1}_{A_{k+1}}(\omega) = 1$

Also, consider $\mathbb{1}_{A_k}(\omega) = 0 \Rightarrow \omega \notin A_k \Rightarrow \omega \in A_k^c$

$\Rightarrow \mathbb{1}_{A_{k+1}}(\omega) = \begin{cases} 1 & \text{if } \omega \in A_{k+1} \cap A_k^c \\ 0 & \text{if } \omega \in A_{k+1}^c \end{cases}$

So, $\mathbb{1}_{A_k} \leq \mathbb{1}_{A_{k+1}}$

Now consider. $x_1 \in A_k$ for atleast one λ.

So. $\forall k \geq \lambda$ $\mathbb{1}_{A_k}(x_1) = 1$ Also. $\mathbb{1}_A(x_1) = 1$

Hence. sequence $\{\mathbb{1}_{A_n}(x_1)\}_n \uparrow \mathbb{1}_A(x_1)$

Now. $x_2 \notin A_k$ for any k

Hence. $\mathbb{1}_{A_k}(x_2) = 0$ $\forall k$

Also $\mathbb{1}_A(x_2) = 0$

So. $\{\mathbb{1}_{A_n}(x_2)\}_n \uparrow \mathbb{1}_A(x_2)$

$\therefore \mathbb{1}_{A_n} \uparrow \mathbb{1}_A$

5.